juliaTDGL 2: a guide to the N-band generalised time-dependent Ginzburg–Landau solver

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Abstract

Release 2.0 (`tdgl2/`, package `TDGL` 2.0.0) generalises the two-band solver of release 1.2 to any number of bands, each with its own Kramer–Watts-Tobin (KWT) relaxation, and adds an open boundary for the self-field, an insulating buffer, Runge–Kutta–Legendre stages and a rotating frame for the bands that KWT makes stiff, and a measuring protocol that brackets an onset by the delay of the first phase slip. With the new terms off it reproduces release 1.2. This guide gives the model, its discretisation, the time step, the boundaries, the protocol, the optional terms, how to run the code and what each check establishes. What is inherited unchanged from 1.2 (the lattice, the link variables, the geometry and its editor) is described in the guide of release 1.2 (`docs/solver_guide.pdf`) and only summarised here.

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1 What the solver computes

The solver integrates the generalised time-dependent Ginzburg–Landau (gTDGL) equations of a superconductor with N order parameters Ψ_μ, coupled by a Josephson matrix, together with the vector potential $\mathbf{A}$, on a three-dimensional box with a buffer of a few cells, under an applied field and a transport current. It was written for the geometric superconducting diode in two-gap materials, where a sample whose shape breaks the mirror symmetry along the direction of vortex motion carries different critical currents $I_c^{\pm}$ in the two directions, and the efficiency is

$$\eta = \frac{I_c^+ - |I_c^-|}{I_c^+ + |I_c^-|}. \tag{1}$$

A run is a current ladder for each polarity. Its output is, at every current, a verdict (silent, dissipating, normal) with the voltage, the number of vortex crossings, the band-resolved currents and the order-parameter amplitudes; the brackets of the onsets and of η; and a machine-readable summary with the units, the code version and the numbers that place the model in its regime (section 2.4).

What release 2.0 adds over 1.2, in one line each:

- N bands and a symmetric Josephson matrix; passive bands (condensate induced by the coupling).
- KWT relaxation per band: the charge imbalance of Gurevich and Vinokur at the contacts, in the gauge $\Phi = 0$.
- First-order Runge–Kutta–Legendre (RKL) stages for the stiff bands and a rotating frame that makes the stationary states with charge imbalance fixed points of the step.
- Self-field boundaries `:bar` (the default) and `:open` (free-space field by a lattice Green function).
- An insulating buffer outside the bar of the sample and its contacts, and a sculpted sample (vacuum inside the box: a cable, a hole), with leads of its own section.
- The onset by the delay of the first phase slip; a verdict for a sample gone normal.
- Runs from a TOML file, materials as files, a JSON summary.

What it does not have: the scalar potential of 1.2 (its Coulomb projection, Thomas–Fermi screening and diffused residual), the `Strict` numerical mode and the explicit default of 1.2. For those, and to reproduce the published results of release 1.x, release 1.2 stays in the repository (`julia/`).

2 The model

2.1 Free energy and units

Lengths are in ξ_1, the coherence length of band 1, time in $t_{\mathrm{GL}} = \xi_1^2/D_1$, fields in H_{c2} of band 1 and currents in J_0. Band 1 has unit stiffness and relaxation, $g_1 = \Gamma_1 = 1$. The functional is

$$F = \sum_\mu \Big[g_\mu |\mathbf{D}\Psi_\mu|^2 - f\,\varepsilon_\mu |\Psi_\mu|^2 + \frac{\beta_\mu}{2}|\Psi_\mu|^4 \Big] - \sum_{\mu<\nu} \gamma_{\mu\nu}(\Psi_\mu^*\Psi_\nu + \text{c.c.}) + \kappa^2 |\nabla\times\mathbf{A}|^2, \qquad (2)$$

with $\mathbf{D} = \nabla - i\mathbf{A}$, $f(\mathbf{r})$ the condensation mask of the geometry (between 0 and 1, or above 1 where the condensate is locally stronger), and $\varepsilon_\mu = a_\mu - T$ (a_μ the value at $T = 0$; with the heat model $T \to \theta(\mathbf{r}, t)$). The sign of $\gamma_{\mu\nu}$ chooses the interband phase (s_{++} for $\gamma > 0$, $s_\pm$ for $\gamma < 0$); with three or more bands and repulsive couplings the phases can be frustrated.

A band is a `Band(m, Γ, a, β, γ̃)` with $g = 1/m$; a model is a `Model` of bands, the matrix γ, κ, the normal conductivity σ_n and the bath temperature T. `two_band(...)` builds the two-band model of release 1.x: $g = (1, 1/m_r)$, $\Gamma = (1, \Gamma_r)$, $a = (1, a_{2n})$, $\beta = 1$, $\sigma_n = 1$, $\tilde\gamma = 0$.

2.2 Equations of motion

For each band, in the gauge $\Phi = 0$,

$$\frac{\Gamma_\mu}{\sqrt{1+\tilde\gamma_\mu^2|\Psi_\mu|^2}}\Big[\partial_t\Psi_\mu + \tilde\gamma_\mu^2\,\Psi_\mu\,\mathrm{Re}(\Psi_\mu^*\partial_t\Psi_\mu)\Big] = g_\mu\mathbf{D}^2\Psi_\mu + (f\varepsilon_\mu - \beta_\mu|\Psi_\mu|^2)\Psi_\mu + \sum_{\nu\neq\mu}\gamma_{\mu\nu}\Psi_\nu, \qquad (3)$$

and for the field

$$\sigma_n(\mathbf{r})\,\partial_t\mathbf{A} = \sum_\mu \mathbf{J}_\mu - \kappa^2\nabla\times\nabla\times\mathbf{A}, \qquad \mathbf{J}_\mu = g_\mu\,\mathrm{Im}(\Psi_\mu^*\mathbf{D}\Psi_\mu), \qquad \mathbf{E} = -\partial_t\mathbf{A}. \qquad (4)$$

With $\tilde\gamma_\mu = 0$ equation (3) is the TDGL of release 1.2. The KWT factor slows the amplitude of a band by $\sqrt{1+\tilde\gamma^2|\Psi|^2}$ relative to its phase; it is the form of Kramer and Watts-Tobin for one band, applied band by band (the extension to several bands is a model choice). $\tilde\gamma_\mu$ multiplies $|\Psi_\mu|$ in the units of the code: a literature value $\gamma_\mu = 2\tau_{E,\mu}\Delta_{\mu,0}/\hbar$ referred to the equilibrium gap becomes $\tilde\gamma_\mu = \gamma_\mu/|\Psi_{\mu,\mathrm{eq}}|$. The material files say so with every number.

Charge imbalance. In the gauge $\Phi = 0$ the electrochemical potential of the condensate is the rate at which its phase turns. Where the injected normal current converts into supercurrent (the conversion layers next to the contacts) the phase of a stationary state turns at that rate, and KWT lengthens the layers. This is the charge imbalance of Gurevich and Vinokur. A band-resolved branch imbalance (Schmid–Schön) is not in the model.

Conservation. The divergence of the field equation gives $\nabla\cdot(\mathbf{J}_s + \sigma_n\mathbf{E}) = 0$, in the continuum and on the lattice (the discrete divergence of a discrete curl vanishes), so no Poisson problem is solved.

2.3 Materials and the normal conductivity

σ_n is given, or that of a dirty BCS material with these bands (`σn = :dirty`, `sigma_dirty`): with the coefficients of Gurevich's two-band theory, $g_\mu\Gamma_\mu/\beta_\mu \propto N_\mu D_\mu$ does not depend on the normalisation of Ψ_μ, and $\sigma_n = \sum_\mu g_\mu\Gamma_\mu/\beta_\mu/5.79$ (one band: $u = \Gamma/\sigma_n = 5.79$, the value of Kramer and Watts-Tobin). The material files in `materials/` carry each parameter with its source and say which numbers are model choices: `mgb2_dirty.toml` and `mgb2_clean.toml` (MgB_2 estimates from the literature), and `mgb2_model_I.toml`, `mgb2_model_II.toml` (the two-band models of the earlier studies, MgB_2-like labels, not quantitative MgB_2).

2.4 The regime numbers

Every result is reported with the numbers that place the model (`regime(md; L)`, also in the summary of a run): the ratio of diffusivities $D_\mu = g_\mu/\Gamma_\mu$ of the last band to the first; the interband Josephson length over the sample length, ℓ_J/L; the invariant $\gamma^2/(\alpha_1\alpha_2)$; $\sigma_n/\sum_\mu g_\mu$; $u_\mu = \Gamma_\mu/\sigma_n$; σ_n over its dirty value; and the conversion length of the normal current at a contact,

$$\lambda_c^2 = \frac{\sigma_n}{\sum_\mu \tilde{\Gamma}_\mu |\Psi_\mu|^2}, \qquad \tilde{\Gamma}_\mu = \frac{\Gamma_\mu}{\sqrt{1+\tilde{\gamma}_\mu^2|\Psi_\mu|^2}}, \tag{5}$$

with L/λ_c. KWT lengthens λ_c: in the two-band notched-box model at $\kappa = 2$, from $0.80\,\xi_1$ to $2.39\,\xi_1$ with $\tilde{\gamma} = 10$. When L is not many λ_c the middle of the sample still carries unconverted normal current, which moves an onset by itself; keep $L/\lambda_c \gtrsim 10$. The pair numbers exist for two bands; for any N the summary has λ_c, u_μ and the two conductivity ratios.

3 Discretisation

3.1 The lattice

As in release 1.2: a staggered Cartesian mesh with Ψ_μ on the nodes, $\mathbf{A}$ on the links as link variables $U = e^{-i\mathbf{A}\cdot d\mathbf{l}}$ (so lattice gauge invariance is exact) and the field on the plaquettes; the condensation mask f on the nodes; covariant Neumann conditions on the six faces of the sample, written as ghost nodes $\Psi_{\text{face}} = U\,\Psi_{\text{interior}}$. Each band is a contiguous array, and every loop over nodes or links is free of reductions, so a run gives the same bits with any number of threads.

3.2 The band update

Without KWT a band is advanced by an explicit Euler step of the right-hand side R_μ of (3), a Jacobi sweep over the interior nodes:

$$\Psi_\mu^{n+1} = \Psi_\mu^n + \frac{dt}{\Gamma_\mu} R_\mu. \tag{6}$$

With KWT the left-hand side is a symmetric positive operator on Ψ seen as a vector of $\mathbb{R}^2$, inverted per node without dividing by Ψ:

$$\partial_t \Psi = R' - \frac{\tilde{\gamma}^2\,\Psi\,\mathrm{Re}(\Psi^* R')}{1+\tilde{\gamma}^2|\Psi|^2}, \qquad R' = \frac{\sqrt{1+\tilde{\gamma}^2|\Psi|^2}}{\Gamma} R. \tag{7}$$

The other bands enter the Josephson term at the start of the step.

3.3 The field update

The curl-curl term is implicit (one solve per step), the supercurrent explicit:

$$\Big(I + \frac{dt}{\sigma_n}\kappa^2 K\Big)\mathbf{A}^{n+1} = \mathbf{A}^n + \frac{dt}{\sigma_n}\mathbf{J}_s^n - \frac{dt}{\sigma_n}\kappa^2 \mathbf{c}, \tag{8}$$

with K the Yee curl-curl and $\mathbf{c}$ the boundary fields. On tensor-product link ranges with $dx = dy = dz$, K is diagonalised by per-axis bases (the singular vectors of the one-dimensional difference), and per wavevector $(I + aK)^{-1} = P_L + P_T/(1 + a|k|^2)$: a direct, exact solve by dense products along each axis. Only the transverse part of $\mathbf{A}$ is implicit; the longitudinal part (the conversion layers) moves with $\mathbf{J}_s$ alone.

Insulating buffer. With σ`vac` the buffer outside the bar of the sample and its contacts has conductivity $\sigma_{\text{vac}} < \sigma_n$, so all of the current crosses the sample, as in a film in vacuum with

leads at its ends. The conductivity per link makes the system non-separable; it is solved by preconditioned conjugate gradients with the separable solve as preconditioner (tolerance 10^{-11}), started from the previous step's solution: consecutive steps differ by little, so it takes 12 iterations where the separable guess takes 27. A ladder starts cold, and what continues a copy of the state (the delay confirmation) gives the solver back, so that a ladder and its resumption keep their bits. The contrast still costs: about 8 times the cost of a step with a metal buffer.

A sculpted sample. A shape element with `vacuum = true` removes the material: no superconductor and no normal metal ($f = 0$ is a normal metal). Its nodes leave the block: $\Psi_\mu = 0$ there, the links that touch one carry no supercurrent ($\tau = 0$), and the band update of a node next to the vacuum reads no neighbour across that link,

$$\Psi_i \leftarrow \Psi_i + \sum_{\pm,\,a} k_a\,\tau_{i,\pm a}\big(U_{i,\pm a}\Psi_{i\pm a} - \Psi_i\big) + \ldots, \tag{9}$$

which is the zero-flux condition of a free surface, the same one the covariant Neumann ghost gives at a face of the box. The insulating buffer follows the sample: a link conducts (σ_n) when both its nodes are in the sample or in a lead, and the leads take the section of the face they touch, so the current enters and leaves through the section of the sample (a cable with leads of its own section). The self-field on the outer faces is the Biot–Savart field of that section. J stays the current density: the sample carries J times its meshed section. A uniform state stays uniform up to the surface to 10^{-13}, and the static current through a cable is J times its section to 2×10^{-3}, none of it through the vacuum. It needs `sigma_vac`; it is not with heat or drag (their stencils do not see the surface); with `qp_exchange` the links to the vacuum are cut (below). Shape files draw figures on any face (`axis = "x"` or `"y"`) and keep the figure with `outside = true`; the shape editor draws on its three views. Example: `examples/cable_k2.toml`.

4 The time step

4.1 Bounds

The step is the smallest of the stability bounds, times 0.99 and `dt_scale` (0.5 by default: no mode of the stiffest band changes sign from one step to the next). For band μ,

$$dt_\mu = \frac{2\Gamma_\mu}{4g_\mu \sum_i dx_i^{-2} + h_\mu} \cdot \frac{1}{\sqrt{1 + \tilde{\gamma}_\mu^2 P_\mu^2}}, \qquad h_\mu = 3\beta_\mu P_\mu^2 - \varepsilon_\mu, \tag{10}$$

where P_μ is the amplitude of the coupled homogeneous equilibrium (interband coupling raises it above $\sqrt{a_\mu/\beta_\mu}$, and a passive band, $a_\mu \le 0$, has one only through the coupling). The KWT factor is the faster phase mode. The explicit supercurrent in Ampère's law gives one more bound, $2\sigma_n / \sum_\mu g_\mu a_\mu/\beta_\mu$, which binds only when $u = \Gamma/\sigma_n$ is large ($u \gtrsim 15$); the explicit curl-curl bound $\sigma_n\, dl/(4\kappa^2)$ disappears with the implicit solve. With the heat model, its explicit diffusion adds $0.15\, dx^2/D_T$. The summary of a run names the bound that set the step.

4.2 Runge–Kutta–Legendre stages

With KWT the band bound falls as $1/\sqrt{1 + \tilde{\gamma}^2 P^2}$ (by 10 with $\tilde{\gamma} = 10$). Instead of shortening the step, the bands are advanced by first-order RKL stages (Meyer, Balsara and Aslam, J. Comput. Phys. 257, 594 (2014)): the step is that of the same model without KWT, and each band takes s stages,

$$Y_0 = \Psi^n, \quad Y_1 = E(Y_0), \quad Y_j = \mu_j E(Y_{j-1}) + \nu_j Y_{j-2}, \quad \mu_j = \frac{2j-1}{j},\ \nu_j = \frac{1-j}{j}, \quad \Psi^{n+1} = Y_s, \tag{11}$$

with E the Euler step of dt/q, $q = s(s+1)/2$, so s stages bridge a ratio of bounds up to q. The linearised band operator is (the KWT factor, symmetric positive per node) times (the Hessian of F): its spectrum is real, which is what the method needs. Stage j sits at the fraction $c_j = j(j+1)/(s(s+1))$ of the step, and its links and ghosts are taken there. `rkl_margin` makes the stages cover the bound times a factor (1.1 by default with KWT): the stages are counted at the equilibrium amplitude P, and an amplitude above it (in the conversion layers) can push the fastest phase mode over the edge. `rkl = false` gives the Euler substeps of release 1.2.

4.3 The rotating frame

In the gauge $\Phi = 0$ a stationary state with charge imbalance is not a fixed point of an explicit step: where its phase turns at rate ω, the step of that rotation, $|1 + i\omega h| > 1$, leaves a radial error $\propto \omega^2 dt$, and KWT, which slows the radial mode and leaves the rotation alone, multiplies it by $\sqrt{1+\tilde{\gamma}^2|\Psi|^2}$. The amplitude in the conversion layers then depends on dt at first order.

With `rotframe` each band is stepped in the gauge that turns with it: $\Psi = e^{i\hat{\omega}\tau}\tilde{\Psi}$, with τ the time from the start of the step and $\hat{\omega}$ the tangential rate of the first stage at each node. The band equation gains $-i\hat{\omega}\tilde{\Psi}$ (tangential, so the KWT factor leaves it alone), the links gain $e^{i(\hat{\omega}_{i+1}-\hat{\omega}_i)\tau}$, and $\Psi^{n+1} = e^{i\hat{\omega}\,dt}\tilde{\Psi}$ turns the state back exactly. A stationary state is then a fixed point of every stage for any dt. Any bounded $\hat{\omega}$ is an exact change of gauge, so the two thresholds only decide where the frame helps: $\hat{\omega} = 0$ where $|\Psi|^2 < 10^{-3}$ (vortex cores, the normal metal), and $|\hat{\omega}|\,dt \le 0.5$. Measured: with the frame the KWT state is the same to six digits at dt, $dt/2$, $dt/4$ and with explicit substeps, for $\tilde{\gamma} = 0$, 10 and 30; $\tilde{\gamma} = 30$ diverged without it.

Defaults. `rotframe` and `rkl_margin` = 1.1 are on when any band has KWT and off otherwise, so every run without KWT keeps its bits.

4.4 Cost

Measured on one core, the two-band notched box ($40 \times 24 \times 6$ cells, $dx = 0.5$, $\kappa = 2$): without KWT 0.078 s per t_{GL}; with $\tilde{\gamma} = 10$ and the defaults $\times 2.2$. The two rotations of a link at each stage (its own and the frame's) are linear in the fraction of the step, so their sum is computed once per band and every stage is one pass with one e^{ia}; the rotating frame then costs nothing over the stages. The explicit step at the KWT bound would be ten times shorter. `dt_scale = 1` halves the cost. On that box the delay of the first slip changed by 0.2 % without KWT (583 against 584 t_{GL}), and with KWT ($\tilde{\gamma} = 10$) the silent state agreed to six digits and the delays at 0.74 and 0.75 were the same: with the rotating frame an onset does not feel it. Flowing voltages were not checked; quote them at the default step.

5 Boundaries of the self-field

The transport current enters as normal current through the two contact faces and its own Ampère field is imposed on the outer faces. Three choices (`selffield_bc`):

`:slab` the boundary of release 1.x: $h_y = \mp H_J$ on the whole z faces. Ampère then encloses the current of the sample and of the buffer, $J\,d\,(W + 2\,b\,dx - dx)$ for a buffer of b cells: 4 % more than the nominal current on the $40 \times 24 \times 6$ box with one cell. It is the choice to reproduce release 1.x.

`:bar` (default) the field of the nominal current spread over the section of the sample, on every outer face, so the sample carries $J\,W\,d$.

`:open` the bar field plus the free-space field of every other current of the box, on all six faces. With $\delta\mathbf{J} = \kappa^2 \nabla_h \times \mathbf{h} - \mathbf{J}_{\mathrm{bar}}$ (a closed current inside the box), $\mathbf{A}_\delta = G * \delta\mathbf{J}/\kappa^2$ and $\mathbf{h}_\delta = \nabla_h \times \mathbf{A}_\delta$ on the face plaquettes, where G is the Green function of the seven-point lattice Laplacian

($-\Delta_h G = \delta$, $G(0) = 0.252731\ldots$), so that the face sees the free continuation of the same discrete scheme. The convolution is Hockney's (zero padding to twice the box, FFT). $\mathbf{h}_\delta$ is recomputed every `open_every` t_{GL} (1 by default) and ramped in between; the iteration contracts (spectral radius at most 0.67, measured in 2D at $\kappa = 2$). It is the field of the screening currents that a mirror face suppresses at small κ. It costs 15–20 % more per t_{GL} than `:bar` on the box above (3 % with KWT), keeps the $J \to -J$ symmetry at $H = 0$ and resumes exactly from a checkpoint.

6 Measuring

6.1 The voltage and the verdict

As in release 1.2: the voltage is the time derivative of the gauge-invariant phase difference $\arg(\Psi_i^* U \Psi_j)$ summed along each measuring line between two planes `delta_v` cells inside the contacts (8 by default), unwrapped line by line, so that each 2π slip of a line is a vortex crossing. A rung integrated for Δt at voltage V carries $|V|\Delta t/2\pi$ crossings per line: it is silent below one, dissipating at ten or more, and undecided in between. The verdict reads band 1, so band 1 must be a band with a condensate of its own, not a passive one.

6.2 A current ladder

Each current starts from the state the previous one left (the order of the ladder is part of the measurement: it gives the hysteresis). A point is integrated in cycles of `lMax` steps until it closes:

`'F'` a fixed point: $|\Psi_\mu|$ and h change by less than the tolerance per cycle after the floor `t_min_GL` (400 t_{GL} by default), and the second half carries fewer than `f_margin` = 0.5 crossings per line (a slow flow near an onset can look fixed for a short cycle).

`'V'` dissipating: after the floor `t_min_GL_res` (1584 t_{GL}), ten or more crossings per line, closed at `v_margin` = 1.1 times that by the end-point estimate so that the slope the verdict uses lands above ten.

`'N'` a sample gone normal: a point that would close `'F'` with $\langle|\Psi_1|^2\rangle$ below `n_frac` = 0.02 of the largest of the `'F'` points before it. The normal state is a fixed point with no phase to cross; with large u or $\tilde{\gamma}$ an onset can be a collapse. `'N'` counts as dissipating.

`'T'` the heat model went above its cut.

`'D'` a point that closed `'F'` and crossed when continued (below).

`bracket_Ic` gives the onset bracket of each polarity from the verdicts, and `run_ladders` the pair and η with its range.

6.3 The onset by the delay

Near a saddle-node onset the delay before the first phase slip, after a jump from a silent state, diverges as

$$D(J) = \frac{a}{\sqrt{J - J_c}} + b. \tag{12}$$

A window of fixed length therefore biases the onset up: the rung just above J_c closes `'F'` on any finite window (in the two-band diode at $\kappa = 2$, delays of 1600–2600 t_{GL} against windows of 400). With `Protocol(t_confirm_GL = ...)`, at the first `'V'` after an `'F'` the `'F'` rungs below it are continued at their current, from copies of their closed states, up to that time in all (at most `confirm_max` = 4): one that crosses becomes `'D'` and the next one down is continued, until one stays silent. `delay_bracket` gives the onset that way; it is the one to quote. `crossing_times(st,`

`s, J; H, T)` measures delays directly from one silent state, and `tools/delay_fit.py` fits the law to two or more of them (with two, $b = 0$; it warns when the fit sits at the edge of its search range).

7 Optional terms

Off by default and meant one at a time (the first three as in release 1.2):

Heat a temperature field above the bath, $\varepsilon_\mu = a_\mu - \theta$ on every node, $\partial_t\theta = D_T\nabla^2\theta + \eta\,[\sum_\mu \Gamma_\mu|\partial_t\Psi_\mu|^2 + \sigma_n|\mathbf{E}|^2] - \gamma(\theta - T)$; the source is the dissipation of the functional, so a Meissner state does not heat. The Joule term is optional (`joule`), and `margin` cells next to the injection faces are left out of the source (contact heat, which leads would carry away). A point above the cut closes `'T'`.

Drag the mixed-gradient term $\nu_{\mu\nu}\mathrm{Re}[(\mathbf{D}\Psi_\mu)^*\cdot\mathbf{D}\Psi_\nu]$ of the functional: band μ feels $\nu_{\mu\nu}\mathbf{D}^2\Psi_\nu$ and the supercurrent carries the cross terms.

Bath temperature a uniform T in $\varepsilon_\mu = a_\mu - T$ for every band.

Two normal fluids (`qp_exchange` $= \nu$, two bands). With one normal current $\sigma_n\mathbf{E}$ the quasiparticles of every band share one potential: their interband exchange is infinitely fast. With a finite rate ν each band has its own normal current $s_\mu\sigma_n(-\nabla\mu_\mu - \partial_t\mathbf{A})$ (s_μ its share of σ_n, `qp_split`, by default the dirty one $\propto g_\mu\Gamma_\mu/\beta_\mu$), and the bands exchange with a conductance $G = \chi_\sigma\nu$. In the gauge $\sum_\mu s_\mu\mu_\mu = 0$ Ampère's law is unchanged and one field is new, $\delta = \mu_\pi - \mu_\sigma$:

$$\sigma_n\nabla^2\delta - G\Big(\frac{1}{s_\sigma} + \frac{1}{s_\pi}\Big)\delta = \frac{C_\pi}{s_\pi} - \frac{C_\sigma}{s_\sigma}, \qquad C_\mu = \mathrm{Im}(\Psi_\mu^* R_\mu), \tag{13}$$

C_μ the conversion of band μ (the divergence of its supercurrent plus the Josephson exchange), with $\delta = 0$ at the contacts and Neumann elsewhere, solved exactly in the eigenbases of the three one-dimensional operators. Each band then turns by $e^{-i\mu_\mu dt}$, tangential, so KWT does not act on it. It is the slaved branch imbalance with a finite exchange (its memory does not move a saddle-node onset); $\nu \to \infty$ is the step of one fluid to $O(1/\nu)$, and in a strip the σ share of the converted current follows the one-dimensional two-fluid law. With N bands `qp_exchange` is the matrix of rates $\nu_{\mu\nu}$ (detailed balance $\chi_\mu\nu_{\mu\nu} = \chi_\nu\nu_{\nu\mu}$): with L the Laplacian of the exchange graph and $S = \mathrm{diag}(s)$, $\sigma_n\nabla^2\mu - S^{-1}L\mu = S^{-1}C - \nmid\!\sum C$, which the eigenvectors of $S^{-1/2}LS^{-1/2}$ split into N Helmholtz problems of the same kind (with two bands, the δ equation above). With the insulating buffer the quasiparticles live in the conductor: δ is solved on the bar of the sample and its contacts, with zero flux to the insulator (separable, exact; the strip then follows the one-dimensional law with the code's contact to 0.004, where the metal buffer, which conducts and exchanges around it, is 0.025 off). In a sculpted sample the links to the vacuum are cut and the equation is no longer separable: conjugate gradients in the inner product with weight 1/2 on each Neumann face (where the mirror-ghost operator is symmetric), preconditioned by the separable solve of the bar and projected on the conductor, started cold every step (20–60 iterations). With the Joule heat the fluids add $\sigma_n\sum_\mu s_\mu|\nabla\mu_\mu|^2 + \sum_{\mu<\nu} G_{\mu\nu}(\mu_\mu - \mu_\nu)^2$ to $\sigma_n|\mathbf{E}|^2$ (the cross term vanishes in the gauge $\sum_\mu s_\mu\mu_\mu = 0$).

Anisotropic bands (`m_axes`, or `m_x`, `m_y`, `m_z` in a material file): a mass per axis in the band update, the supercurrent and the step bound ($4\sum_i g_i/dx_i^2$). An isotropic band keeps its bits; a band with $m_x = 4$ at $dx = 0.5$ is the isotropic one at $dx = 1$ to 10^{-9}.

Thermal noise (`noise` $= \tilde{T}$, `noise_seed`): a Langevin term $\langle\eta_\mu\eta_\mu^*\rangle = 2\Gamma_\mu\tilde{T}\delta(\mathbf{r} - \mathbf{r}')\delta(t - t')$ on every band, so that the discrete system is distributed as $e^{-F/\tilde{T}}$ (checked: mean, variance and

correlation of decoupled nodes, and the spatial covariance against the linear theory of the lattice to 0.1–1 %). Seeded, saved in the checkpoints (resuming keeps the bits). With KWT the mobility of a node depends on Ψ, $M = (s/\Gamma)I - (\tilde{\gamma}^2/\Gamma s)\psi\psi^{\mathsf{T}}$ with $s = \sqrt{1+\tilde{\gamma}^2|\Psi|^2}$: the kick is scaled apart along Ψ ($1/\Gamma s$) and across it (s/Γ), and the Itô drift $\tilde{T}\nabla \cdot M/(2dV) = -\tilde{T}\tilde{\gamma}^2(1+s^2)\psi/(2\Gamma s^3 dV)$ keeps $e^{-F/\tilde{T}}$ (checked: the same mean and variance as without KWT, the amplitude s times slower and the phase s times faster; without the drift the mean of $|\Psi|^2$ comes out 1.39 instead of 1). With `noise_field` the same bath acts on the field: Ampère is $\sigma_l \partial_t \mathbf{A} = -\frac{1}{2}\delta F/\delta \mathbf{A}$ per unit volume, so a link takes $\Delta A_l = \sqrt{\tilde{T}dt/(\sigma_l dV)}\, n$, added to the explicit part of the field step before the implicit solve. In a normal box the field energy of the free (transverse) modes is $\sum_m (\tilde{T}/2)/(1+\lambda_m dt/2)$ to 0.03 %, with σ_n and with the insulating buffer: the implicit Euler of the field leaves the stiff mesh modes colder and the slow ones right to $O(\lambda dt)$.

None of them sets the step in practice (the heat diffusion bound, $0.15\, dx^2/D_T$, is 0.15 at $dx = 0.5$ and $D_T = 0.25$, against $dt \approx 0.02$), so an implicit treatment would gain nothing.

8 Running

8.1 From a file

A run is one TOML file:

```
[material]   file = "mgb2_model_II.toml"      (or kappa, sigma_n, T, gamma12 | gamma,
                                                 and [[material.band]] tables inline)
[mesh]       nx, ny, nz, dx, buf
[geometry]   GEOM_MODE, geom_file, ...         (the shape keys of release 1.2)
[scheme]     the Stepper keywords: dt_scale, rkl, rotframe, rkl_margin,
             selffield_bc ("bar" | "open" | "slab"), open_every, sigma_vac, ...
[protocol]   t_min_GL, t_min_GL_res, t_confirm_GL, ...
[heat]       D_T, gamma_bath, eta, joule, margin       (optional; drag goes in [scheme])
[run]        J = [...], H, out_dir
```

and one command:

```
OPENBLAS_NUM_THREADS=1 julia --project=tdgl2 \
    tdgl2/scripts/run.jl RUN.toml [--dir OUT]
```

It writes into the run folder the file it was given, `output/rungs_J+.csv` and `rungs_J-.csv` (one line per rung: verdict, voltage, crossings, band currents and amplitudes, delays), `restart/` (JLD2 checkpoints, from which a ladder resumes), and `output/summary.json`: the units, the code version and commit, the scheme as resolved (step, the bound that set it, stages, `dt_scale`, `rotframe`, `rkl_margin`, the boundary), the protocol, the regime numbers, and the brackets of the window and of the delay with η. **Stopping and resuming.** The checkpoint of every closed point holds, in one file written at once, the state, the open boundary and the temperature field, the solution of the insulating solve, and all the ladder carries from one point to the next (the convergence references, the reference of the 'N' verdict, the closed rungs waiting for the delay confirmation). `STOP.txt` in the run folder stops a run after the current point; `--resume` continues it, and the result is the one of the run without a stop, to the bit (the suite checks it, with the delay confirmation half done and with the open boundary). A killed run loses only the point it was on. `--polarity +` and `--polarity -` run the two polarities as two processes into the same folder; the second to end writes the summary, the same as one process running both. A folder that already holds a ladder is not run over without `--resume`.

Two examples are in `examples/`: `wedge_k25.toml` (minutes) and `kwt_k2.toml` (about an hour; the onset with KWT, $\tilde{\gamma} = 10$, rises to $(0.73, 0.74]$ and $(0.75, 0.76]$ with $\eta = -1.34$ %); and one

per option: `open_k2`, `insulating_k2`, `heat_k25` (heat with Joule, a contact margin and drag) and `three_bands`.

8.2 From Julia

```
using TDGL
md = two_band(xi_ratio=4.0, κ=2.0, gamma_kwt=10.0)
gr = Grid(40, 24, 6; dx=0.5)
mk = build_masks(GeomSpec(GEOM_MODE="WEDGE"), gr)
st = Stepper(md, gr, mk)                 # production scheme; rotframe on (KWT)
L  = ladder(st, initial_state(md, gr, mk, 0.03), [0.0, 0.5, 0.7, 0.74]; H=0.03,
            pr=Protocol(t_confirm_GL=3000.0, stop_after_V=1))
delay_bracket(L)
```

8.3 The terminal menu, campaigns and tools

The terminal menu of release 1.2 (`julia/scripts/tdgl-tui.sh`) lists the runs of 2.0 that have an `out_dir`, with their points, and launches one from a run file (key `n`; with two CPUs, one process per polarity), stops it after the current point (`s`) and resumes it (`u`). Its launch wizard (`l`) ends with the choice of release: `2` writes the run file of release 2 (the two-band model of 1.2, KWT off) and launches it. For campaigns, one single-threaded process per physical core (`OPENBLAS_NUM_THREADS=1`); on a laptop with hybrid cores the total gain saturates near 2.5× one process (memory bandwidth). The Slurm scripts of release 1.2 (`examples/`) serve with their command replaced by `tdgl2/scripts/run.jl RUN.toml --dir OUT`. `tools/ladder_sheet.py` draws the standard sheet of one or more ladders (crossings per line, $|V|$, time per rung and the current through the sample against $|J|$).

9 Checks

The suite (`tdgl2/test/run.jl`, a few minutes) checks:

- **Release 1.2.** The step, the masks of every catalogue shape, the checkpoints and a two-polarity ladder point by point, with the new terms off and `selffield_bc = :slab`, against states written by release 1.2. To the bit with OpenBLAS's Haswell kernels and Julia code for that CPU (`TDGL_TEST_BITS=1 OPENBLAS_CORETYPE=Haswell julia -C haswell`, any x86 machine with AVX2); to 10^{-10} elsewhere, where the dense products of the separable solve and the SIMD sums round differently (measured: 10^{-15} on the states, 10^{-12} on ladder voltages, which are time averages). Verdicts and counts are exact either way.
- **Energy.** The discrete balance $dF/dt = -D+W$ on a two-band box relaxing from a random state, where W is the work of the supercurrent on the links between two face nodes (about 0.5 % of D there; with it the residual is $O(dt)$). With those links off, $W = 0$ and F decreases at every step: an exact gradient flow.
- **The rotating frame.** The KWT state at dt and $dt/2$ and with substeps, to 10^{-5}.
- **Three bands.** Equal repulsive couplings frustrate the phases: the ground state breaks time reversal with phases 0, $\pm 2\pi/3$ and $|\Psi|^2 = 1 + |\gamma|$, the analytic values, to 10^{-6}.
- **Symmetry.** A sample with the mirror $y \to -y$ at $H \neq 0$: the state at $+J$ is the mirror of the state at $-J$ to rounding (10^{-15}), with each boundary, so $\eta = 0$.
- **The open boundary.** The lattice Green function, $\mathbf{h}_\delta = 0$ without currents, a closed source, the nominal current through the sample to 2×10^{-3}, exact resumption, the $J \to -J$ symmetry at $H = 0$.

- **The protocol.** The verdicts `'N'` and `'D'`, `crossing_times`, `delay_bracket`, the regime numbers (one band against the conversion law).

Against pyTDGL (version 0.9; one band in 2D, the one-band equation of this code with $\Gamma = u$, $\sigma_n = 1$, κ large; pyTDGL's current unit is four times the GL unit at its interface, and it solves in the GL unit inside): the depairing current of a narrow strip, 0.385; the conversion length with KWT for $\tilde{\gamma} = 0$–30, 0.2 % between the codes and 0.1 % from $\lambda_c^2 = \sqrt{1 + \tilde{\gamma}^2|\Psi|^2}/(u|\Psi|^2)$; the onset of a notched strip, independent of $\tilde{\gamma}$ in both codes, which with the mesh approaches pyTDGL's value (8 % below it at $dx = 0.5\,\xi$, 4 % at $0.25\,\xi$: the $O(dx)$ error at the notch tip). The flowing voltage has not been compared.

10 Limitations

- **Absolute onsets carry an $O(dx)$ error** of 5–12 % at $dx = 0.5\,\xi$, at any κ: the tip of a notch is resolved at the scale of a cell, and at small κ the face rows of the sample, Neumann copies without an equation of their own, carry a large share of the screening supercurrent (51 % at $\kappa = 2$, $dx = 0.5$; 30 % at 0.25). Ratios between polarities (η) and trends do not feel it; quote absolute onsets with their mesh.
- **With KWT the conversion at the contacts is long**: keep $L/\lambda_c \gtrsim 10$.
- **The silent state is driven.** The conversion layers carry a steady field even when the order parameter is static, so an onset depends on the relaxation rates and on the contact, not only on the free energy. Quote onsets for the rates, the KWT parameters and the boundary used.
- **The charge imbalance is slaved**: KWT per band is the branch (Schmid–Schön) imbalance in its slaved form, and `qp_exchange` adds a finite interband exchange; its memory and the scalar potential of 1.2 are not in. The normal fluids depend on the share of σ_n between the bands, a model choice.
- **The insulating buffer costs about 8 times per t_{GL}** what a metal buffer costs: its preconditioner does not see the contrast between the bar and the buffer. A sculpted sample needs it, so it costs the same.
- **Anisotropy** is a mass per axis and band; σ_n and drag stay isotropic.
- **The implicit solve needs $dx = dy = dz$.**
- **No GPU**: the loops are per node and per link with no reductions, so a port is mechanical; it has not been done.
- **One single-threaded process per physical core** for campaigns. Within one run, threads give the same bits but no speed on a laptop (measured: ×0.8–1.06 with 2–8 threads).